XIAMEN UNIVERSITY
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ASSIGNMENT 2



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1. An abrupt silicon pn junction at $T = 300$ K is uniformly doped with $N_a = 5 \times 10^{17}$ cm^{-3} and $N_d = 8 \times 10^{15}$ cm^{-3}. The cross-sectional area of the pn junction is 9×10^{-4} cm^2. An inductance is placed in parallel with the pn junction. (*a*) With a reverse-bias voltage of $V_R = 14$ V applied to the pn junction, the resonant frequency of the circuit is $f = 2.25$ MHz. What is the value of the inductance? (*b*) Using the results of part (*a*), what is the resonant frequency if the reverse-bias voltage is (*i*) $V_R = 18$ V and (*ii*) $V_R = 53$ V?

2. A silicon p^+ n junction diode is to be designed to have a breakdown voltage of at least 60 V and to have a forward-bias current of $I_D = 50$ mA while still operating under low injection. The minority carrier lifetimes are $\tau_0 = \tau_{n0} = \tau_{p0} = 4 \times 10^{-7}$ s. Determine the doping concentrations and the minimum cross-sectional area. ($N_d \cong 8 \times 10^{15}$ cm^{-3} Let $N_a = 5 \times 10^{17}$ cm^{-3}, D_p=11 cm^2/s)

3. (*a*) Consider a silicon sample at $T = 300$ K doped with 10^{16} cm^{-3} donor atoms. Let $\tau_{p0} = 7 \times 10^{-7}$ s. A light source turns on at $t = 0$ producing excess carriers with a uniform generation rate of $g' = 3 \times 10^{20}$ cm^{-3} s^{-1}. At $t = 4 \times 10^{-7}$ s, the light source turns off. (*i*) Derive the expression(s) for the excess carrier concentration as a function of time over the range $0 \le t \le \infty$. (*ii*) What is the value of the excess concentration when the light source turns off. (*b*) Repeat part (*a*) for the case when the light source turns off at $t = 3 \times 10^{-6}$ s.

4. An ideal n-channel MOSFET has the following parameters: $V_T = 0.45$ V, $\mu_n = 427$ cm^2/V-s, $t_{ox} = 12$ nm = 120 Å, $W = 25$ μm, and $L = 1.5$ μm. (*a*) calculate I_D and V_{DS} for $0 \le V_{DS} \le 3$ V and for $V_{GS} = 0$, 0.6, 1.2, 1.8, and 2.4 V. calculate V_{DS} (*sat*) point. (*b*) compute the I_D and V_{GS} for $0 \le V_{GS} \le 2.4$ V and for $V_{DS} = 0.1$ V.

5. A constant electric field, E = 15 V/cm, exists in the +*x* direction of an n-type gallium arsenide semiconductor for $0 \le x \le 60\mu$m. The total current density is a constant and is $J = 600$ A/cm^2. At $x = 0$, the drift and diffusion currents are equal. Let $T = 300$ K and $\mu_n = 8800$ cm^2/V-s. (*a*) Determine the expression for the electron concentration $n(x)$. (*b*) Calculate the electron concentration at $x = 0$ and at $x = 60\mu$m. (*c*) Calculate the drift and diffusion current densities at $x = 60\mu$m.

6. The total junction capacitance of a GaAs pn junction at $T = 300$ K is found to be 2.10 pF at $V_R = 2$ V. The doping concentration in one region is measured and found to be 8.2×10^{16} cm^{-3}, and the built-in potential is found to be $V_{bi} = 1.30$ V. Determine (*a*) the doping in the other region of the pn junction and (*b*) the cross-sectional area. (*c*) The reverse-bias voltage is changed and the capacitance is found to be 0.80 pF. What is the value of V_R?

7. Chung wants to know about n-channel enhancement-mode MOSFET with threshold voltage 0.40 V, t_{ox} = 22 nm = 220 Å, L = 2.4 μm, and W = 15 μm. (*a*) Assuming a constant mobility of μ_n = 485 cm^2/V-s, calculate I_D for $V_{GS} - V_T$ = 2.0 V when biased at (*i*) V_{DS} = 0.7 V, (*ii*) V_{DS} = 1.0 V, (*iii*) V_{DS} = 1.25 V, and (*iv*) V_{DS} = 2.0 V. (*b*) Consider the piecewise linear model of the carrier velocity versus V_{DS} shown in Figure 1. Calculate I_D for the same voltage values given in part (*a*). (*c*) Determine the V_{DS} (sat) values for parts (*a*) and (*b*).

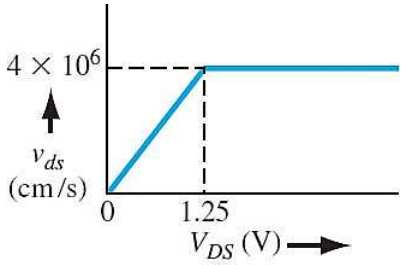


Figure 1.

8. A Schottky diode and a pn junction diode have cross-sectional areas of $A = 8 \times 10^{-4}$ cm^2. The reverse-saturation current densities at T = 300 K of the Schottky diode and pn junction are 3×10^{-8} A/cm^2 and 2×10^{-12} A/cm^2, respectively. A forward-bias current of 0.9 mA is required in each diode. (*a*) Determine the forward-bias voltage required across each diode. (*b*) If the voltage from part (*a*) is maintained across each diode, determine the current in each diode if the temperature is increased to 500 K. (Take into account the temperature dependence of the reverse saturation currents. Assume E_g = 1.14 eV for the pn junction diode and ϕ_{B0} =0.84 V for the Schottky diode.)

9. Rahman is interested to study about Metal-semiconductor junctions. He found out there is a junction name MS (Metal-Silicon). He also likes to have few assumptions based on his findings;

1. Work function $\Phi_M = 4.60$ eV.
2. Electron affinity of Si $\chi = 4.03\ eV$

a) Now let's say we have N-type silicon that has $N_D = 10^{17} cm^{-3}$ draw an energy band diagram. Indicate the Schottky barrier height on your drawing (explain all steps).
b) Now assume the silicon is is P-type with $N_A = 10^{17} cm^{-3}$ and draw an energy band diagram. Indicate the Schottky barrier height on your drawing.
c) Explain why each of the above two examples is, or is not, expected to be a rectifying junction.
d) Compute the Schottky barrier height for each part a and b.

10. A MOS capacitor is made with a silicon substrate doped with $N_a = 5 \times 10^{17}\text{cm}^{-3}$ of boron, 2nm of silicon dioxide, and an n+ polysilicon gate doped such that $Ef = Ec +$ 0.05 eV. Assume there are no significant oxide charges. Determine the charge on the gate, the voltage dropped across the oxide and the voltage dropped across the silicon with the following voltages applied between the gate and the substrate:
(a) $V_{gb} = -1$ V;
(b) $V_{gb} = 0{:}0$ V;
(c) $V_{gb} = 1$ V
Sketch the charge densities, electric fields and energy band diagrams in each case. What are the capacitances at low and high frequencies in each of the above cases?

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